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# 1 Printer Maintenance Manual

**A practical guide to keeping your office printers reliable, efficient, and long-lasting.**

## 1.1 Welcome

This manual provides standardized procedures for the daily operation, maintenance, and troubleshooting of office printers. Consistently following these guidelines helps reduce downtime, extend hardware lifespan, improve print quality, and lower the total cost of ownership.

> **Audience**
>
> This document is intended for IT support staff, facility managers, and designated printer operators. End users may also refer to the **Paper Handling** and **Troubleshooting** sections for common issues.

## 1.2 How to Use This Manual

| Section | Purpose | When to Use |
|---|---|---|
| Getting Started | Printer overview, components, and first-time setup | Onboarding a new operator or device |
| Daily Checklist | Routine inspection tasks | Every morning / end of day |
| Cleaning & Care | Step-by-step cleaning procedures | Weekly or when print quality drops |
| Consumables | Toner, ink, drum, and waste toner replacement | When supply alerts appear |
| Paper Handling | Loading paper and resolving jams | Paper jam or misfeed errors |
| Troubleshooting | Symptom-to-solution reference | Printer malfunction or error codes |
| Safety | PPE, electrical, and thermal safety | Before any maintenance task |
| FAQ | Quick answers to common questions | General inquiries |

## 1.3 Key Principles

1. **Preventive over reactive** — routine maintenance costs far less than emergency repairs.
2. **Clean hands, clean machine** — always power off and unplug before internal cleaning.
3. **Use approved supplies** — third-party toner and paper can cause damage and void warranties.
4. **Log every action** — record maintenance in the service log for trend analysis.
5. **When in doubt, escalate** — do not disassemble components beyond this manual's scope.

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## 1.4 Service Log Template

Record each maintenance event in the printer service log:

| Date | Device ID | Performed by | Action | Parts replaced | Meter reading | Next due |
|---|---|---|---|---|---|---|
| | | | | | | |

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*Last updated: 2026-09-10*

# 2 Getting Started

## 2.1 Overview

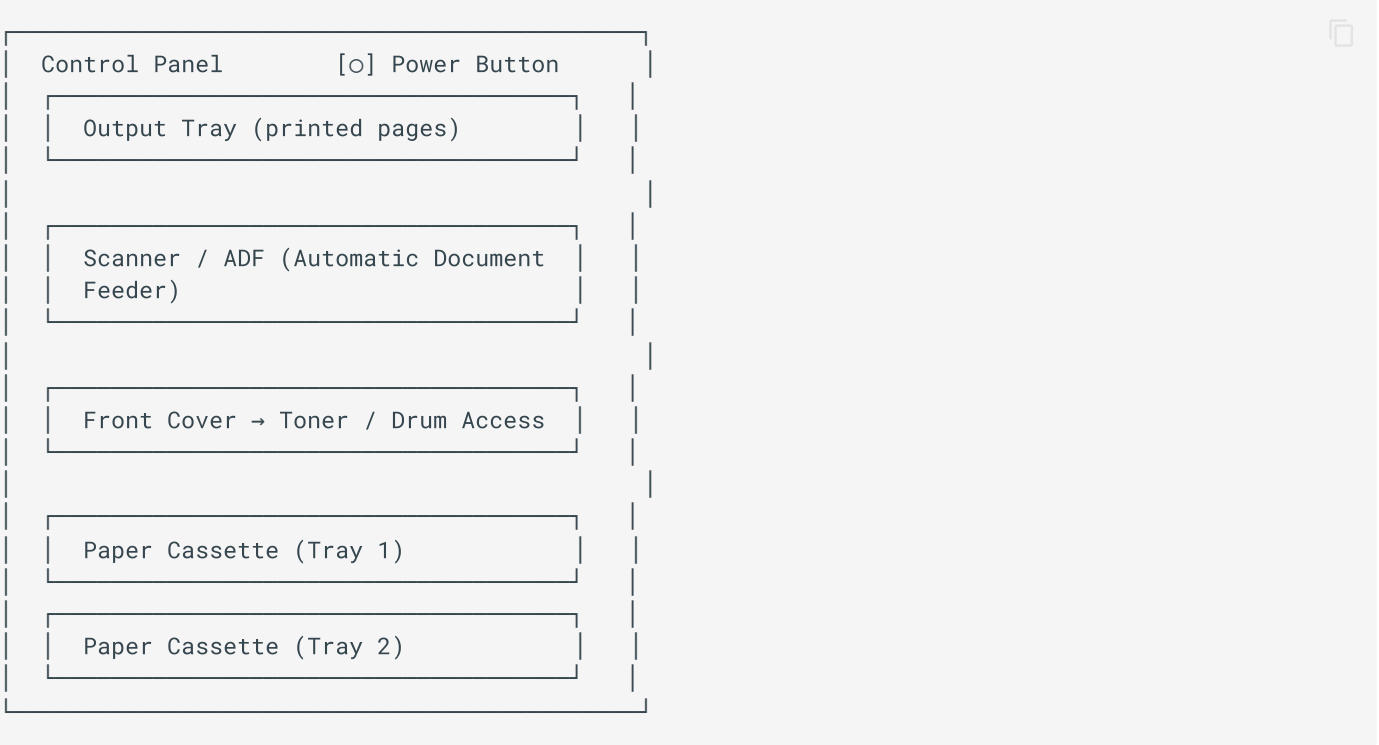
Before performing any maintenance, familiarize yourself with the printer's main components, control panel indicators, and the tools required for routine tasks. This section prepares you for the procedures described throughout this manual.

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## 2.2 Printer Components

Understanding the physical layout is essential for safe and efficient maintenance.

### 2.2.1 Front View




### 2.2.2 Rear View

- **Rear cover** — access for paper jam removal in the fuser area
- **Power port** — always disconnect before internal maintenance
- **USB / Ethernet ports** — network and local connections
- **Fuser unit** — gets very hot; allow 30 minutes to cool before touching

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## 2.3 Control Panel Indicators

| Indicator | Color | Meaning | Action |
|---|---|---|---|
| Power LED | Green solid | Printer is on and ready | — |
| Power LED | Green blinking | Processing / receiving data | Wait |
| Attention LED | Amber solid | Cover open, paper jam, or toner low | Check display message |
| Attention LED | Amber blinking | Error requiring intervention | See Troubleshooting |
| Toner LED | Amber | Toner cartridge low or empty | Replace toner |
| Error LED | Red | Fatal error or hardware fault | Power cycle; escalate if persists |

## 2.4 Required Tools and Supplies

Keep the following items available at all times:

- Lint-free, static-free microfiber cloths
- Isopropyl alcohol (90%+ concentration)
- Compressed air can (or anti-static vacuum)
- Cotton swabs (for narrow areas)
- Latex or nitrile gloves
- Replacement toner / ink cartridges (genuine OEM)
- Printer service log sheet
- Flashlight (for inspecting internal paper paths)

> ⚠ **Do not use**
>
> - Paper towels or facial tissue (leave lint and can scratch)
> - Water-based cleaners on electrical components
> - Ammonia or abrasive cleaners on the platen glass
> - Vacuum cleaners without anti-static protection (can damage electronics)

## 2.5 First-Time Setup Checklist

When a new printer is deployed, complete the following before releasing it to users:

1. **Unbox and inspect** — verify all accessories are present and there is no shipping damage.
2. **Remove all shipping locks** — pull out all orange/red tape, protective clips, and drum locks.
3. **Place on a stable surface** — allow at least 10 cm clearance on all sides for ventilation.
4. **Install initial toner and drum** — follow the included quick-start guide.
5. **Load paper** — fan the paper stack before loading; do not exceed the tray capacity line.
6. **Connect power and network** — use a dedicated power outlet; avoid daisy-chained power strips.
7. **Power on and configure** — set date/time, network IP, and admin password.
8. **Print a configuration page** — verify connectivity and record the device serial number and IP address.
9. **Update firmware** — check the manufacturer's support site for the latest firmware.
10. **Label the printer** — affix a label with device ID, location, and support contact.

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## 2.6 User Roles and Responsibilities

| Role | Responsibilities |
|---|---|
| **End User** | Load paper, clear simple jams, report issues, avoid foreign objects |
| **Printer Operator** | Daily checklist, routine cleaning, consumable replacement, basic troubleshooting |
| **IT Support** | Network configuration, firmware updates, advanced troubleshooting, vendor escalation |
| **Facilities** | Power outlet maintenance, environmental control (temperature/humidity), physical security |

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## 2.7 Environmental Requirements

Maintain the following conditions for optimal printer performance:

- **Temperature**: 10 °C – 32.5 °C (50 °F – 90 °F)
- **Humidity**: 20% – 80% RH (non-condensing)
- **Avoid**: direct sunlight, dust, proximity to air conditioners or heaters, and strong vibration
- **Power**: stable AC supply with proper grounding; use a surge protector

**Pro tip**

Keep printers away from carpeted areas if possible. Static electricity from carpet can cause paper feed issues and damage sensitive components.

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*Next: Daily Checklist*

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# 3 Daily Maintenance Checklist

## 3.1 Overview

Consistent daily and weekly checks are the most effective way to prevent printer failures and maintain print quality. This checklist should be performed by the designated printer operator at the start and end of each business day.

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## 3.2 Daily Morning Checklist

Perform these tasks before the first print job of the day.

### 3.2.1 1. Visual Inspection

- [ ] Ensure the printer power is on and the Ready indicator is solid green
- [ ] Check for no error or warning messages on the control panel display
- [ ] Confirm the output tray is empty and ready to receive pages
- [ ] Listen for no unusual noises (grinding, clicking, squealing) during warm-up
- [ ] Inspect for no paper scraps or foreign objects visible around the paper path
- [ ] Verify cables are securely connected and not damaged

### 3.2.2 2. Paper Supply

- [ ] Load paper cassettes with the correct paper type and size
- [ ] Fan paper before loading to prevent double-feeds
- [ ] Adjust paper guides snugly against the paper stack (not too tight)
- [ ] Ensure paper level is above the minimum fill line
- [ ] Check for no curled, wrinkled, or damp paper in the trays
- [ ] Store paper in a cool, dry location (original packaging until use)

### 3.2.3 3. Consumable Status

- [ ] Check toner/ink level is above 20% (order replacement if below)
- [ ] Verify drum unit life is above 10% (schedule replacement if near end)
- [ ] Ensure waste toner container is not full
- [ ] Load staples (if multifunction finisher) as needed

- [ ] Confirm spare consumables are available on-site

### 3.2.4 4. Print Quality Test

- [ ] Print a test page from the control panel
- [ ] Inspect for no streaks, smudges, or faded areas on the test page
- [ ] Confirm text is sharp and legible at normal reading distance
- [ ] Verify colors are accurate (for color printers)
- [ ] Check for no paper jams during the test print

> **Record the meter reading**
>
> Every morning, record the total page count (meter reading) in the service log. This helps track usage patterns and schedule preventive maintenance.

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## 3.3 Daily End-of-Day Checklist

Perform these tasks after the last print job of the day.

- [ ] Empty the output tray and file pages
- [ ] Clear any pending or stuck print jobs in the queue
- [ ] Ensure the scanner bed/ADF is clear of original documents
- [ ] Close the printer cover fully
- [ ] Keep the work area around the printer clean and tidy
- [ ] Enable power-saving mode (or power off per policy)

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## 3.4 Weekly Checklist

Perform these tasks every Friday (or the last business day of the week).

### 3.4.1 Cleaning

- [ ] Wipe the exterior casing with a dry or slightly damp lint-free cloth
- [ ] Clean the scanner glass and ADF feed rollers
- [ ] Clean the paper feed rollers with isopropyl alcohol
- [ ] Remove dust from ventilation slots with compressed air

- [ ] Clean the control panel display with a screen-safe wipe

### 3.4.2 Functional Checks

- [ ] Print a configuration/supplies status page
- [ ] Verify all paper trays are recognized correctly
- [ ] Test scan-to-email and scan-to-folder functions (if applicable)
- [ ] Test duplex (double-sided) printing
- [ ] Check for firmware updates and schedule installation
- [ ] Review the service log for recurring issues

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## 3.5 Monthly Checklist

- [ ] Perform a full deep clean (see Cleaning & Care)
- [ ] Inspect the fuser unit for wear or damage (allow to cool fully)
- [ ] Check and clean the transfer belt/transfer roller
- [ ] Verify the printer's IP address and network configuration
- [ ] Review usage statistics and reorder consumables proactively
- [ ] Verify the printer is covered under warranty or service contract
- [ ] Update the asset inventory record

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## 3.6 Quarterly / Preventive Maintenance

- [ ] Replace the fuser unit (if within maintenance kit schedule)
- [ ] Replace transfer roller and pickup rollers (maintenance kit)
- [ ] Clean or replace the ozone filter
- [ ] Lubricate moving parts per manufacturer specification
- [ ] Run a full diagnostic test from the service menu
- [ ] Calibrate color registration (color printers)

> ⚠ **Qualified technician only**
>
> Fuser replacement, transfer belt replacement, and internal lubrication should be performed by a qualified service technician. Do not attempt these tasks unless trained.

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## 3.7 Checklist Quick Reference

| Frequency | Key Tasks | Est. Time |
|---|---|---|
| **Daily (AM)** | Visual check, paper, consumables, test print | 5 min |
| **Daily (PM)** | Empty tray, clear queue, tidy area | 3 min |
| **Weekly** | Exterior clean, scanner clean, functional tests | 15 min |
| **Monthly** | Deep clean, fuser inspection, network check | 30 min |
| **Quarterly** | Maintenance kit replacement, full diagnostic | 60 min (by technician) |

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## 3.8 Escalation Triggers

Stop routine maintenance and escalate to IT support if any of the following occur:

- Recurring paper jams (more than 2 per day) after cleaning
- Persistent print quality issues after toner replacement and cleaning
- Unusual odors (burning, ozone) or smoke
- Repeated error codes that do not clear after power cycle
- Physical damage to the printer or accessories
- Network connectivity issues that cannot be resolved locally

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*Next: Cleaning & Care*

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# 4 Cleaning and Care

## 4.1 Overview

Regular cleaning helps prevent common printer problems such as paper jams, streaky prints, feed errors, and premature component wear. This section provides all cleaning procedures, from basic exterior wiping to internal component care.

**Before you begin**

1. Power off the printer using the power button.
2. Unplug the power cable from the wall outlet.
3. Wait at least 30 minutes for the fuser to cool completely.
4. Wear nitrile gloves to avoid skin oils on components.
5. Never touch the drum (photosensitive) surface with bare fingers.

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## 4.2 Exterior Cleaning

### 4.2.1 Frequency: Weekly

1. Power off and unplug the printer.
2. Wipe the exterior casing, control panel, and output tray with a dry lint-free cloth.
3. For stubborn marks, dampen the cloth slightly with water or a mild, non-abrasive cleaner.
4. Clean the control panel touchscreen with a dedicated screen wipe (no alcohol on touchscreens).
5. Dust the ventilation slots with compressed air (hold can upright, 15 cm away).
6. Plug back in and power on.

**Never spray directly**

Never spray cleaning solution directly onto the printer. Always apply to the cloth first, and ensure the cloth is damp, not dripping.

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## 4.3 Scanner Glass and ADF Cleaning

### 4.3.1 Frequency: Weekly or when streaks appear on scans

### 4.3.2 Flatbed Scanner Glass

1. Open the scanner lid.
2. Spray a small amount of glass cleaner onto a lint-free cloth (not directly on the glass).
3. Wipe the glass in a circular motion, then buff dry with a clean cloth.
4. Inspect under good lighting — any dust or smudge will appear as a line on scanned pages.
5. Clean the white reference strip on the underside of the lid as well.

### 4.3.3 Automatic Document Feeder (ADF)

1. Open the ADF cover.
2. Dampen a lint-free cloth with isopropyl alcohol.
3. Clean the ADF feed roller, separation pad, and glass strip.
4. Rotate the roller manually to clean the entire circumference.
5. Allow all components to dry completely (5 minutes) before closing.

> **Diagnostic trick**
>
> If scanned pages have a vertical line in the same position every time, the cause is almost always dirt on the ADF glass strip, not the flatbed glass. Clean the narrow glass strip first.

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## 4.4 Paper Feed Roller Cleaning

### 4.4.1 Frequency: Monthly or when misfeeds increase

Dirty feed rollers are the #1 cause of paper jams and multi-feeds.

1. Power off, unplug, and open the paper cassette.
2. Locate the feed roller inside the printer, above where the paper sits.
3. Dampen a lint-free cloth or cotton swab with isopropyl alcohol.
4. Gently wipe the roller surface, rotating it to clean all sides.
5. If the roller surface is smooth or shiny (glazed), it may need replacement — contact IT support.
6. Allow to dry for 5 minutes.
7. Reinstall the cassette and print a test page.

### 4.4.2 Pickup Roller (Tray 2 and above)

Repeat the same procedure for each paper tray's pickup roller. Access may require removing the tray and looking into the cavity.

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## 4.5 Internal Paper Path Cleaning

### 4.5.1 Frequency: Monthly or after clearing a bad jam

1. Power off, unplug, and wait 30 minutes for the fuser to cool.
2. Open the front cover and remove the toner / drum unit.
3. Use compressed air to gently blow dust and paper dust from the paper path.
   - Hold the can upright, 15–20 cm away.
   - Use short bursts; prolonged spraying can cause the can to freeze.
4. Wipe accessible rubber rollers with an alcohol-dampened cloth.
5. Inspect for any torn paper scraps left behind from previous jams.
6. Reinstall the toner / drum unit and close all covers.
7. Plug in, power on, and print a test page.

> ⚠ **Do not touch**
>
> - **Drum unit surface** — fingerprints cause permanent print defects
> - **Fuser rollers** — even after 30 minutes, may still be warm
> - **Transfer belt** — extremely delicate; use only compressed air, never touch

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## 4.6 Toner / Ink Spill Cleanup

If toner powder spills during cartridge replacement:

1. Do not use a standard vacuum — The toner is conductive and can damage the vacuum and create a fire risk.
2. Use a toner-specific vacuum (HEPA-filtered, anti-static) if available. Otherwise, use a dry lint-free cloth to gently lift the powder.
3. For residual dust, use a slightly damp cloth (water only).
4. Allow all surfaces to dry completely before closing the printer.
5. Dispose of contaminated cloths per local regulations.

**Toner and heat**

Never use a hot air blower or heat gun on spilled toner. Toner melts at approximately 180 °C and will permanently bond to surfaces.

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## 4.7 Fuser Area Cleaning

### 4.7.1 Frequency: Quarterly (by qualified personnel)

The fuser uses heat and pressure to bond toner to paper. It requires careful handling.

1. Ensure the printer has been powered off and unplugged for at least 30 minutes.
2. Open the rear or side fuser access cover.
3. Visually inspect the fuser rollers for toner buildup, scratches, or wear.
4. Use a dry lint-free cloth to gently wipe excess toner from the fuser entrance and exit guides.
5. **Do not** touch the hot roller surface — it is coated and easily damaged.
6. Close the cover and run a few blank pages to clean residual toner.

**Fuser replacement**

If the fuser shows visible scratches, film tears, or causes repeated print defects, it must be replaced. This is a wear item covered in the periodic maintenance kit.

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## 4.8 Cleaning Schedule Summary

| Component | Frequency | Method |
|---|---|---|
| Exterior casing | Weekly | Dry / damp lint-free cloth |
| Scanner glass | Weekly | Glass cleaner + lint-free cloth |
| ADF rollers / glass strip | Weekly | Isopropyl alcohol |
| Feed rollers | Monthly | Isopropyl alcohol, rotate fully |
| Internal paper path | Monthly | Compressed air + cloth |

| Component | Frequency | Method |
|---|---|---|
| Fuser area | Quarterly | Dry cloth, visual inspection |
| Ventilation slots | Weekly | Compressed air |
| Control panel | Weekly | Screen-safe wipe |

## 4.9 What Not to Clean

| Component | Why |
|---|---|
| Drum unit (photosensitive cylinder) | Fingerprints and light cause permanent damage |
| Transfer belt | Extremely delicate; contact causes scratches |
| Fuser hot roller | Coating damage; risk of burns |
| Laser scanner unit | Sealed unit; opening voids warranty and can misalign laser |
| Print head (inkjet) | Use manufacturer's head cleaning utility only |

*Next: Consumables*

# 5 Consumables Management

## 5.1 Overview

Consumables include toner cartridges, ink cartridges, drum units, waste toner containers, maintenance kits, and paper. Proper management helps maintain print quality, reduces the risk of unexpected downtime, and extends the useful life of each component.

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## 5.2 Consumable Lifecycle

```
Order → Receive → Store → Install → Monitor → Replace → Dispose
  │                                                  │
  └──────────────── Reorder point ──────────────────┘
```

### 5.2.1 Reorder Triggers

| Consumable | Reorder When | Recommended Stock |
|---|---|---|
| Toner / ink cartridge | Level drops below 20% | 1 spare per printer model |
| Drum unit | Life remaining below 15% | 1 spare for high-volume printers |
| Waste toner container | 80% full indicator | 1 spare |
| Maintenance kit | 80% of rated page count | Schedule service visit |
| Paper | Below 2 weeks of supply | 1 month supply |

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## 5.3 Toner Cartridge Replacement

### 5.3.1 When to Replace

- The control panel displays "Toner Low" or "Toner Empty"
- Printouts appear faded or show light vertical streaks
- The toner LED is lit amber

### 5.3.2 Procedure (Laser Printer)

1. Prepare the replacement cartridge and keep it ready for installation. Shake it gently 5–6 times from side to side to distribute the toner evenly.
2. Ensure the printer is powered on so the cartridge can seat correctly.
3. Open the front cover fully.
4. Remove the old cartridge by gripping the handle and pulling it straight out.
5. Prepare the new cartridge by removing the protective seal along the indicated tab. Do not touch the drum surface.
6. Insert the new cartridge, aligning it with the rails and pushing firmly until it clicks into place.
7. Close the front cover. The printer will run a calibration cycle for 15–60 seconds.
8. Print a test page to confirm print quality.
9. Dispose of the old cartridge through the manufacturer's recycling program.

**Extend toner life**

When the "Toner Low" message first appears, remove the cartridge, shake it gently side to side five times, and reinstall it. This can yield an additional 50–200 pages and provides time to arrange a replacement.

### 5.3.3 Toner Storage

- Store in a cool, dry place (10–30 °C, 30–70% RH)
- Keep in the original packaging until ready to install
- Do not store near heat sources or in direct sunlight
- Avoid high humidity, because toner absorbs moisture and can clump
- Typical shelf life is 2–3 years from the manufacture date; check the packaging for details

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## 5.4 Drum Unit Replacement

### 5.4.1 What Is a Drum?

The drum is the photosensitive cylinder that transfers the toner image onto the paper. It is a wear item with a rated lifespan of approximately 12,000–60,000 pages, depending on the model.

### 5.4.2 When to Replace

- The control panel displays "Drum Life End" or "Replace Drum"
- Printouts show repeating spots or smudges at regular intervals; the interval typically matches the drum circumference

- Faded or ghost images persist even after installing a new toner cartridge

### 5.4.3 Procedure

1. Power on the printer and open the front cover.
2. Remove the toner cartridge if it is attached to the drum unit.
3. Pull the drum unit assembly straight out to remove it.
4. Unpack the new drum unit and remove all protective covers.
5. Insert the new drum unit until it clicks into place.
6. Reinstall the toner cartridge into the new drum unit.
7. Close the front cover.
8. Reset the drum counter. This step is critical; the printer will not recognize the new drum unless the counter is reset from the control panel menu.
9. Print a test page.

**Reset the counter**

Forgetting to reset the drum counter is the most common mistake. The printer will continue to display "Replace Drum" and may stop printing. Refer to the device's service manual for the exact reset procedure.

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## 5.5 Waste Toner Container

### 5.5.1 When to Replace

- The display shows "Waste Toner Full" or "Waste Toner Box Near Full"
- Print quality declines because of toner buildup

### 5.5.2 Procedure

1. Power off and unplug the printer.
2. Open the front or side access cover, depending on the printer model.
3. Carefully pull out the waste toner container while keeping it level to avoid spills.
4. Seal the old container in a plastic bag.
5. Install the new container and push until it clicks into place.
6. Close the cover, plug the printer back in, and power it on.
7. Reset the waste toner counter if prompted.

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# 5.6 Ink Cartridge Replacement (Inkjet Printers)

## 5.6.1 When to Replace

- The display shows the ink level is low or empty
- Printouts have missing colors or horizontal white lines
- The ink droplet icon is lit

## 5.6.2 Procedure

1. Power on the printer.
2. Open the ink cartridge access door so the carriage moves to the center.
3. Wait for the carriage to stop moving.
4. Press down on the empty cartridge to release it, then pull it out.
5. Remove the new cartridge from its packaging and peel off the protective tape covering the nozzles.
   - Do not touch the copper contacts or the nozzle plate.
6. Insert the new cartridge into the correct slot, matching the color labels, and push it up until it clicks.
7. Close the access door.
8. The printer will run an automatic priming cycle.
9. Print a nozzle check pattern to verify that all colors are firing correctly.
10. If lines are missing, run the head cleaning utility for 1–2 cycles, then perform a deep clean if necessary.

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# 5.7 Paper Selection and Storage

## 5.7.1 Recommended Paper Specifications

| Attribute | Recommendation |
|---|---|
| Weight | 75–90 g/m² (20–24 lb) for general use |
| Brightness | 90+ ISO for crisp text |
| Moisture content | 4–6% (too dry = static and jams; too damp = curl and wrinkles) |
| Finish | Smooth, uncoated paper for laser printers |
| Sizes | A4, Letter, Legal (check printer specifications) |

### 5.7.2 Paper Storage Best Practices

- Store paper in its original ream wrapper until use
- Keep paper flat on a shelf rather than on the floor, because floor moisture can cause curling
- Maintain 30–50% relative humidity in the storage area
- Do not open multiple reams at once
- Fan the paper stack before loading to separate sheets and reduce static
- For long-term storage, keep paper in a sealed cabinet with desiccant

### 5.7.3 Paper Types to Avoid

- Paper that has already been printed on, unless using manual feed for a specific media type
- Heavily textured or rough paper, which can damage the rollers
- Paper with clips, staples, or adhesive notes
- Damp or wrinkled paper
- Labels or transparencies that are not rated for laser printers, because they can melt in the fuser

**Laser vs. Inkjet media**

Never use inkjet-specific transparency film or labels in a laser printer. The heat can melt the adhesive and cause severe jams. Always use media rated for the correct printer type.

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## 5.8 Maintenance Kit

A maintenance kit typically includes:

- Fuser unit
- Transfer roller
- Pickup rollers for all trays
- Separation pads
- Feed rollers

### 5.8.1 Replacement Interval

Maintenance kits are typically replaced every 100,000–200,000 pages, depending on the printer model. Check the kit life counter in the printer configuration page.

**Professional installation recommended**

Installing a maintenance kit often requires disassembling multiple parts of the printer. Schedule replacement with an authorized service provider to avoid damage and preserve warranty coverage.

## 5.9 Disposal and Recycling

| Item | Disposal Method |
|---|---|
| Toner / ink cartridges | Use the manufacturer take-back program (HP, Canon, Brother, and others typically offer free recycling) |
| Waste toner | Seal in a plastic bag and dispose of it as electronic waste; do not place it in regular trash |
| Drum units | Manufacturer recycling or an e-waste facility |
| Old paper | Recycle through a standard paper recycling stream |
| Packaging | Cardboard and plastic recycling |

## 5.10 Consumable Inventory Log

Track consumable stock using this template:

| Item | Model / Part # | Printer Model | In Stock | On Order | Reorder Level | Last Ordered |
|---|---|---|---|---|---|---|
| Toner (Black) | | | | | | |
| Toner (Cyan) | | | | | | |
| Drum Unit | | | | | | |
| Waste Toner | | | | | | |

| Item | Model / Part # | Printer Model | In Stock | On Order | Reorder Level | Last Ordered |
|---|---|---|---|---|---|---|
| Paper (A4) | | | | | | |

*Next: Paper Handling*

# 6 Paper Handling and Jam Resolution

## 6.1 Overview

Paper jams are the most common printer issue. Most jams are caused by improper paper loading, worn rollers, or damaged paper. This section covers correct paper loading procedures and step-by-step jam removal for all common jam locations.

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## 6.2 Loading Paper Correctly

### 6.2.1 Standard Procedure

1. Remove the paper cassette from the printer by pulling it straight out.
2. Fan the paper stack — hold the stack at both ends and flex it, then fan the pages. This separates sheets and reduces static.
3. Tap the stack on a flat surface to align the edges neatly.
4. Load the paper into the cassette with the print side facing down (check the cassette label for orientation).
5. Adjust the paper guides — slide the side and end guides until they touch the paper stack snugly.
   - Too loose → paper skews → jams
   - Too tight → paper buckles → jams
6. Do not overfill — keep the paper below the maximum fill line marked on the cassette.
7. Reinsert the cassette firmly until it clicks.
8. Confirm paper size on the control panel if prompted.

> **The golden rule**
>
> If the printer has a paper size sensor, always verify that the displayed size matches the actual paper. A mismatch causes 90% of "phantom" jams and misfeeds.

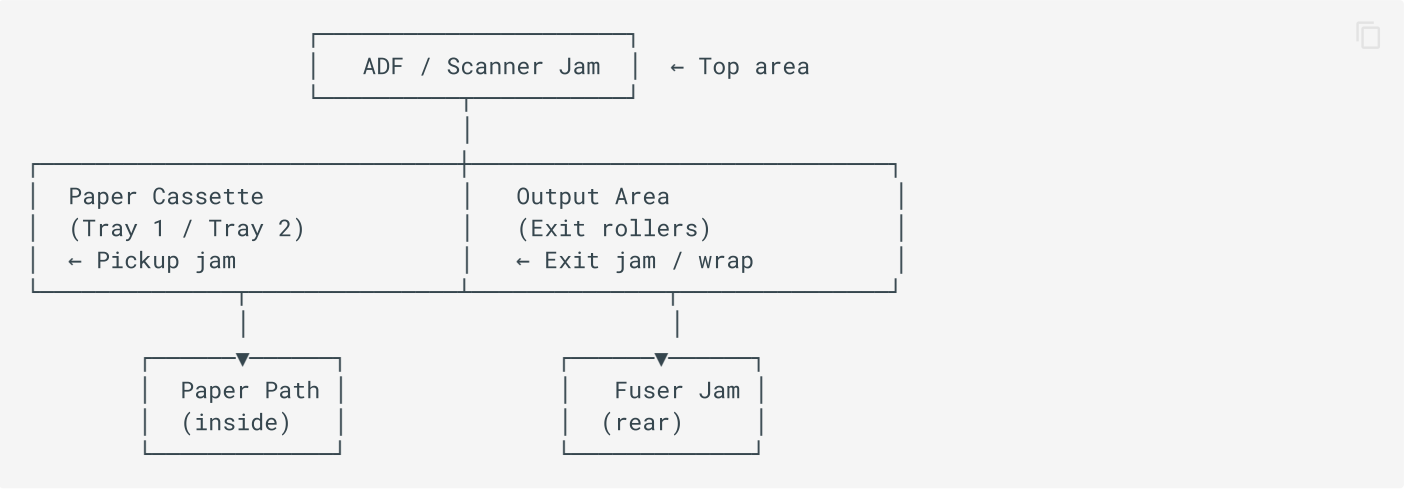
### 6.2.2 Loading Envelopes, Labels, and Special Media

| Media Type | Tray | Orientation | Notes |
|---|---|---|---|
| Envelopes | Manual feed (MP tray) | Face down, flap side up, left edge first | Do not use envelopes with windows or metal clasps |

| Media Type | Tray | Orientation | Notes |
|---|---|---|---|
| Labels | Manual feed or Tray 1 | Face down, label side down | Use only laser-rated labels; do not feed partially used label sheets |
| Transparencies | Manual feed | Rough side down (check packaging) | Use only laser-rated transparencies |
| Cardstock / Heavy paper | Manual feed | Face down | Check maximum weight specification (typically 163–216 g/m²) |
| Letterhead / Preprinted | Cassette or MP tray | Face down, top edge toward the printer | Test one sheet first to confirm orientation |

## 6.3 Understanding Jam Locations

Paper jams can occur in several areas. The printer's control panel usually displays a diagram showing the jam location.




## 6.4 General Jam Removal Procedure

**Safety first**

1. Power off the printer before opening covers.
2. The fuser area is extremely hot — wait 30 minutes if accessing the rear.
3. Pull paper slowly and steadily in the direction of normal paper travel. Never yank.
4. If the paper tears, remove every scrap — even a tiny piece will cause another jam.

### 6.4.1 Step-by-Step

1. Stop the printer — press the Stop/Cancel button, then power off.
2. Check the display — note the indicated jam location.
3. Open the appropriate cover for the jam location (see sections below).
4. Remove the paper gently, using both hands if possible.
5. Inspect for torn pieces — use a flashlight to check rollers and guides.
6. Close all covers firmly.
7. Power on — the printer may run a calibration or automatically reprint the jammed page.
8. Print a test page to confirm normal operation.

---

## 6.5 Jam Location: Paper Cassette / Pickup Area

### 6.5.1 Symptoms

- "No paper" error even when paper is loaded
- Paper does not feed at all
- Grinding sound from the cassette area
- Multiple sheets feed at once

### 6.5.2 Resolution

1. Remove the paper cassette.
2. Check for a misfed sheet partially visible at the top of the cassette cavity.
3. If visible, pull the paper straight down and out gently.
4. If not visible, check the pickup roller area — paper may be wrapped around the roller.
5. Remove the paper, then inspect the pickup roller:
   - Clean with isopropyl alcohol if dusty (see Cleaning).
   - If the roller surface is smooth/glazed, it needs replacement.

6. Fan and reload the paper, ensuring guides are correctly adjusted.
7. Reinsert the cassette and test.

### 6.5.3 Preventing Pickup Jams

- Do not overload the cassette (stay below the fill line)
- Fan paper before loading
- Keep paper dry (humidity causes sheets to stick)
- Clean feed rollers monthly
- Do not mix paper sizes in the same cassette

---

## 6.6 Jam Location: Internal Paper Path

### 6.6.1 Symptoms

- "Paper jam inside" or "Jam in print path" message
- Paper stops halfway through the printer

### 6.6.2 Resolution

1. Open the front cover.
2. Remove the toner / drum unit assembly.
3. Look inside the paper path for the jammed sheet.
4. Pull the paper straight out in the direction of normal travel (toward the front or output, depending on model).
5. If the paper is wrapped around a roller, rotate the roller manually (in the normal direction) to feed the paper out.
6. Use tweezers for small torn pieces if necessary.
7. Reinstall the toner / drum unit.
8. Close the cover and test.

---

## 6.7 Jam Location: Fuser Area (Rear)

**HOT — 30 minute cool-down required**

The fuser operates at approximately 180–220 °C. Always power off, unplug, and wait at least 30 minutes before opening the rear fuser cover.

### 6.7.1 Symptoms

- "Jam at rear" or "Fuser jam" message
- Paper exits partially, then stops
- Crinkled or wrapped paper at the output

### 6.7.2 Resolution

1. Power off, unplug, and wait 30 minutes.
2. Open the rear cover (or fuser access door).
3. Lower the fuser pressure release levers (if present) — this releases pressure on the rollers.
4. Pull the paper straight out toward the rear.
   - If the paper is wrapped around the upper fuser roller, rotate the roller gear manually to feed it out.
5. Inspect for torn scraps inside the fuser.
6. Return the pressure levers to their original position.
7. Close the rear cover.
8. Plug in, power on, and print 2–3 blank pages to clear residual toner.

> ⚠ **Do not use sharp objects**
>
> Never use scissors, knives, or screwdrivers to remove paper from the fuser. Scratching the fuser roller causes permanent print defects.

---

## 6.8 Jam Location: Output Area

### 6.8.1 Symptoms

- Paper exits partially and stops
- "Jam at output" message
- Paper is curled or folded at the exit

### 6.8.2 Resolution

1. Open the front or top output cover.
2. Gently pull the paper straight out toward the output tray.
3. If the paper is torn, open the rear cover and remove remaining pieces from the fuser exit.
4. Check the output rollers for wear or toner buildup.

5. Close covers and test.

---

## 6.9 Jam Location: ADF / Scanner

### 6.9.1 Symptoms

- "Document jam" in the ADF
- Original document does not feed or stops halfway

### 6.9.2 Resolution

1. Open the ADF cover.
2. Gently pull the original document straight out.
3. If torn, remove all scraps from the feed path.
4. Clean the ADF feed roller and separation pad with isopropyl alcohol.
5. Close the ADF cover.
6. Test with a single sheet.

### 6.9.3 Preventing ADF Jams

- Remove staples, paper clips, and sticky notes before loading
- Do not load more sheets than the ADF capacity (usually 20–50 pages)
- Fan originals before loading
- Do not use wrinkled, torn, or heavily curled originals
- Adjust the ADF paper guides to the document width

---

## 6.10 Recurring Jams: Root Cause Analysis

If the same printer jams repeatedly (2+ times per day), investigate these causes:

| Symptom Pattern | Likely Cause | Fix |
|---|---|---|
| Jams at pickup, every time | Worn/glazed pickup roller | Clean or replace roller |
| Jams at same point in path | Torn paper scrap stuck in path | Full inspection, remove all scraps |
| Jams only with certain paper | Paper too damp, too thick, or wrong type | Replace paper stock |

| Symptom Pattern | Likely Cause | Fix |
|---|---|---|
| Jams after toner replacement | Toner cartridge not seated correctly | Remove and reinstall firmly |
| Jams at fuser repeatedly | Worn fuser rollers or pressure pads | Replace fuser (maintenance kit) |
| Multi-feeds (2+ sheets at once) | Static buildup or worn separation pad | Fan paper, replace separation pad |
| Jams only in duplex mode | Duplex path roller worn | Clean/replace duplex rollers |

## 6.11 After Clearing a Jam

Always perform these steps after any jam:

- [ ] All covers are fully closed and latched
- [ ] No torn paper remains inside (use flashlight)
- [ ] Paper cassette is reinserted firmly
- [ ] Printer displays "Ready" (no error)
- [ ] Test page prints successfully
- [ ] Print quality is normal (no streaks from toner spill)
- [ ] Record the jam in the service log (date, location, cause)

---

*Next: Troubleshooting*

# 7 Troubleshooting Guide

## 7.1 Overview

This section provides symptom-to-solution reference for the most common printer problems. Problems are organized by category. Always start with the simplest solution, such as a power cycle, and progress to more complex fixes.

---

## 7.2 Quick Fix: Power Cycle

Before any advanced troubleshooting, perform a power cycle. This resolves approximately 40% of transient errors.

1. Press the power button to turn off the printer.
2. **Unplug** the power cable from the wall outlet (not only the printer).
3. Wait **60 seconds** — this allows capacitors to discharge and memory to clear.
4. While waiting, check that all covers are closed and no paper is jammed.
5. Plug the power cable back in.
6. Power on and wait for the printer to reach the "Ready" state.
7. Print a test page.

> **Network printers**
>
> If the printer is network-connected, also restart the network switch or router port if connectivity issues persist.

---

## 7.3 Print Quality Problems

### 7.3.1 Faded or Light Prints

| Possible Cause | Solution |
|---|---|
| Toner cartridge low or empty | Replace toner cartridge |
| Toner is unevenly distributed | Remove cartridge, shake gently 5 times side to side, reinstall |
| Economode / toner save mode is on | Disable toner save mode in printer settings |

| Possible Cause | Solution |
|---|---|
| Drum unit near end of life | Replace drum unit |
| Paper is damp or too thick | Replace with fresh, dry paper within spec |
| Transfer roller dirty or worn | Clean transfer roller; replace if needed |
| Laser scanner window dirty | Clean the scanner window (accessible via front cover on some models) |

### 7.3.2 Vertical Streaks or Lines

| Pattern | Cause | Solution |
|---|---|---|
| Black vertical line | Damaged drum or toner cartridge | Replace toner; if persists, replace drum |
| White vertical line | Toner clog or dirty developer roller | Replace toner cartridge |
| Repeating spots at regular intervals | Drum damage or fuser roller damage | Measure interval; replace drum or fuser |
| Streaks only on scans/copies | Dirty scanner glass or ADF glass strip | Clean scanner glass (see Cleaning) |
| Streaks in the same position on all prints | Dirty corona wire or primary charge roller | Clean per manufacturer instructions; replace drum |

### 7.3.3 Smudges or Toner Rubs Off

| Cause | Solution |
|---|---|
| Fuser temperature too low | Check paper type setting (thick paper requires higher fuser temp) |
| Fuser unit worn or failing | Replace fuser (maintenance kit) |
| Paper is too thick or has wrong texture | Use paper within printer's weight specification |

| Cause | Solution |
|---|---|
| Toner is non-OEM / incompatible | Use genuine OEM toner |
| Print speed too high for media | Reduce print speed in driver settings for heavy media |

### 7.3.4 Ghosting / Repeated Images

A faint duplicate of the image appears further down the page.

- **Cause**: Worn drum or fuser roller; the image is not fully transferred or fused.
- **Fix**: Replace the drum unit first. If ghosting persists, replace the fuser.

### 7.3.5 Color Issues (Color Printers)

| Symptom | Cause | Solution |
|---|---|---|
| One color missing | Empty cartridge or clogged nozzle | Replace cartridge; run head cleaning (inkjet) |
| Colors are wrong / shifted | Color registration misaligned | Run automatic color calibration from the menu |
| Color banding | Dirty LED print head or faulty cartridge | Clean; replace affected cartridge |
| All colors faded | Transfer belt nearing end of life | Replace transfer belt (maintenance kit) |
| Background haze | Incorrect toner density setting | Reset to factory defaults; replace drum |

---

## 7.4 Paper Feed Problems

### 7.4.1 Printer Says "No Paper" But Paper Is Loaded

1. Remove the cassette and inspect the pickup roller.
2. Clean the roller with isopropyl alcohol.
3. Check that the paper size sensor flag is not stuck.
4. Verify the paper is below the maximum fill line and above the minimum.
5. Ensure the cassette is fully inserted (push firmly until it clicks).

6. If the problem persists, the pickup roller may need replacement.

### 7.4.2 Multiple Sheets Feed at Once

1. Remove the paper and fan it thoroughly to reduce static.
2. Check that the paper is not damp (replace with fresh stock).
3. Clean the separation pad with isopropyl alcohol.
4. If the separation pad is worn smooth, replace it.
5. Do not mix paper types or weights in the same cassette.

### 7.4.3 Paper Skews / Prints Crooked

1. Check that the paper guides in the cassette are snug against the stack.
2. Ensure the paper is loaded squarely (not at an angle).
3. Clean the feed rollers — uneven roller wear causes skew.
4. Check for a torn paper scrap in the paper path causing asymmetric drag.
5. Verify the paper cassette is not warped or damaged.

---

## 7.5 Error Codes

### 7.5.1 Common Error Codes (Generic)

| Code | Meaning | Action |
|---|---|---|
| **E1** / **13.xx** | Paper jam | Clear jam per Paper Handling |
| **E2** / **10.xx** | Paper size mismatch | Verify cassette paper size matches driver setting |
| **E3** / **50.xx** | Fuser error | Power cycle; if persists, replace fuser |
| **E4** / **51.xx** | Laser scanner error | Power cycle; if persists, service required |
| **E5** / **52.xx** | Motor error | Check for obstructions; service required |
| **E6** / **53.xx** | Memory error | Power cycle; reseat memory if upgradeable |
| **E7** / **54.xx** | Color registration error | Run auto calibration; service if persists |

| Code | Meaning | Action |
|---|---|---|
| **79.xx** | Firmware error | Power cycle; update firmware; remove faulty print jobs |
| **50.4** | Fuser power supply error | Check power outlet; try different circuit |

**Model-specific codes**

Error codes vary by manufacturer (HP, Canon, Brother, Epson, Xerox, etc.). Always refer to the printer's service manual for exact code definitions. The codes above are generic patterns.

### 7.5.2 79 Service Error (HP Common)

This is a firmware crash, often caused by a corrupted print job.

1. Power off the printer and unplug for 60 seconds.
2. While off, clear all print jobs from the computer's print queue.
3. Plug in and power on.
4. If the error returns without sending a job, update the firmware.
5. If the error appears only when printing a specific file, the file is corrupted — re-save or re-export it.

---

## 7.6 Network and Connectivity Problems

### 7.6.1 Printer Offline / Not Found on Network

1. Print a configuration page from the printer's control panel to verify its IP address.
2. From a computer on the same network, ping the printer's IP address.
   - If ping fails: check network cable, switch port, and printer network settings.
   - If ping succeeds: the issue is with the printer driver/port on the computer.
3. Verify the printer has a static IP address (DHCP addresses can change and break the connection).
4. Restart the print spooler service on Windows:

```
net stop spooler
net start spooler
```

5. Remove and re-add the printer on the affected computer.
6. Update the printer driver to the latest version.

### 7.6.2 Slow Printing Over Network

- Check network cable (replace Cat5e with Cat6 if possible)
- Reduce print resolution or use "Fast" print mode for drafts
- Disable "Keep printed documents" in the printer driver
- Update firmware and network card drivers
- For large documents, use a direct USB connection as a workaround

### 7.6.3 Scan-to-Email Fails

1. Verify the printer has a valid DNS server configured.
2. Check SMTP server settings (address, port, authentication, encryption).
3. Test with a known-working email address.
4. If using Office 365 / Gmail, ensure "less secure app" access or app passwords are configured.
5. Check firewall settings — the printer must be able to reach the SMTP port.

---

## 7.7 Hardware and Mechanical Problems

### 7.7.1 Unusual Noises

| Noise | Likely Cause | Action |
|---|---|---|
| Grinding / clicking | Gear train issue, torn paper in gears | Inspect for paper scraps; service if persists |
| Squealing | Dry or worn rollers | Clean/lubricate per service manual; replace rollers |
| High-pitched whine | Fuser bearing or motor failing | Replace fuser or motor |
| Loud thumping | Imbalanced drum or toner cartridge | Reseat or replace cartridge |
| Buzzing / humming | Power supply or transformer | Normal if faint; service if loud or new |

### 7.7.2 Printer Does Not Power On

1. Verify the power cable is firmly connected at both ends.
2. Try a different power outlet (test the outlet with another device).
3. Try a different power cable.

4. Check the power switch is in the "On" position.
5. If the printer has a power supply module, it may need replacement.
6. If there was a recent power surge, the fuse or power supply may be damaged — service required.

### 7.7.3 Control Panel Display Issues

- **Blank display**: Check power; adjust display contrast/brightness; the display module may be faulty.
- **Display shows garbled characters**: Firmware corruption — update firmware; if unable, service required.
- **Touchscreen unresponsive**: Clean the screen; recalibrate touch if available; the touch panel may need replacement.
- **Buttons not responding**: Sticky or worn buttons — clean around buttons; control panel may need replacement.

---

## 7.8 Software and Driver Problems

### 7.8.1 Print Jobs Stuck in Queue

1. Open the print queue on the computer.
2. Cancel all documents.
3. Restart the print spooler service (Windows) or cups service (Linux/macOS).
4. Power cycle the printer.
5. Resend the print job.

### 7.8.2 Wrong Paper Size or Orientation

- Check the paper size setting in the application's print dialog.
- Verify the printer driver's default paper size matches the loaded paper.
- Ensure the printer's control panel reports the correct paper size for each tray.
- For mixed-size documents, use "Select by Paper Source" or auto-select.

### 7.8.3 Printer Prints Garbled Characters

- The wrong printer driver is installed — download and install the correct driver for the model.
- The print job is corrupted — cancel and resend.
- The interface cable is faulty — replace USB or network cable.
- Firmware is outdated — update to the latest version.

---

## 7.9 When to Escalate

Contact IT support or an authorized service provider when:

- Error codes persist after power cycle and basic troubleshooting
- Print quality does not improve after toner, drum, and cleaning
- Recurring jams (2+ per day) after roller cleaning
- Any hardware error (fuser, laser scanner, motor, power supply)
- Physical damage to the printer or accessories
- The printer is under warranty — do not open sealed components (voids warranty)
- Firmware update fails or bricks the device
- Network issues that cannot be resolved with basic checks

---

## 7.10 Troubleshooting Decision Flow

```
Start
  │
  ├─ Power cycle resolves it? ──Yes──→ Done (log transient error)
  │
  └─ No
      │
      ├─ Error code on display? ──Yes──→ Look up code → Follow fix
      │
      └─ No code
          │
          ├─ Paper jam? ──Yes──→ Clear jam → Test
          │
          ├─ Print quality issue? ──Yes──→ Check toner → Drum → Clean → Fuser
          │
          ├─ Connectivity issue? ──Yes──→ Check IP → Ping → Driver → Spooler
          │
          └─ Other ──→ Consult service manual → Escalate
```

---

*Next: Safety*

---

# 8 Safety Guidelines

## 8.1 Overview

Printer maintenance involves electrical, thermal, mechanical, and chemical hazards. Following these safety guidelines protects both personnel and equipment. All maintenance personnel must read and understand this section before performing any procedure in this manual.

---

## 8.2 General Safety Principles

1. **Stop if in doubt.** If a procedure feels unsafe or beyond your training, do not proceed — escalate to a qualified technician.
2. **One person, one task.** Do not perform maintenance while another person is also working on the same printer.
3. **Communicate.** If you are working on a printer, place an "Out of Service — Maintenance in Progress" sign so users do not attempt to use it.
4. **No shortcuts.** Never bypass safety interlocks or tape down door switches to operate the printer with covers open.
5. **First aid.** Know the location of the nearest first aid kit, eyewash station, and fire extinguisher.

---

## 8.3 Electrical Safety

### 8.3.1 Before Any Internal Maintenance

- [ ] Power off the printer using the power button (do not unplug the printer while it is running)
- [ ] Unplug the power cable from the wall outlet
- [ ] If the printer has a separate power switch, set it to "Off"
- [ ] Verify the printer is disconnected by attempting to power it on

### 8.3.2 Electrical Hazards

| Hazard | Risk | Precaution |
|---|---|---|
| High voltage power supply | Electric shock, burns | Never open the power supply enclosure; service only by qualified personnel |

| Hazard | Risk | Precaution |
|---|---|---|
| Laser scanner unit | Laser radiation can damage eyes | Do not open the laser scanner cover; interlocks prevent exposure when closed |
| Wet hands / environment | Increased shock risk | Ensure hands are dry; do not use water near electrical components |
| Damaged power cable | Shock, fire | Inspect cables regularly; replace if cracked, frayed, or pinched |
| Overloaded outlet | Fire | Use a dedicated outlet; do not daisy-chain power strips |

**Laser safety**

Laser printers use a laser beam to write the image on the drum. Never attempt to defeat the door interlock switches or look into the laser scanner assembly. The laser is invisible and can cause permanent eye damage.

---

## 8.4 Thermal Safety

### 8.4.1 Fuser Unit

The fuser is the hottest component in a laser printer, operating at **180–220 °C (356–428 °F)**.

- Always wait 30 minutes after powering off before touching the fuser area.
- Even after 30 minutes, the fuser may still be warm — touch cautiously.
- Use the designated handles or levers; never grab the fuser rollers directly.
- If you smell burning or see smoke, power off immediately, unplug, and evacuate the area if necessary.

### 8.4.2 Other Hot Components

- **Transfer roller** — can become warm during long print runs
- **Duplexer path** — paper passing through may be hot to the touch
- **Inkjet print head** — some inkjet printers heat the print head to 40–60 °C

### 8.4.3 Burn First Aid

1. Remove the heat source immediately.

2. Cool the burn under cool (not cold) running water for 10–20 minutes.
3. Do not apply ice, butter, or ointments to severe burns.
4. Cover with a clean, non-stick dressing.
5. Seek medical attention for burns larger than a coin, blistering burns, or burns on the face/hands.

---

# 8.5 Mechanical Safety

## 8.5.1 Pinch Points and Moving Parts

Printers contain gears, rollers, belts, and motors that can pinch fingers or pull in jewelry, hair, or clothing.

- **Never** reach into the printer while it is powered on and operating.
- Remove rings, watches, bracelets, and dangling jewelry before maintenance.
- Tie back long hair.
- Do not wear loose sleeves or scarves near moving parts.
- Use the provided levers and handles — do not force components.

## 8.5.2 Heavy Lifting

- Desktop printers (under 20 kg): lift with your legs, not your back.
- Floor-standing / multifunction printers (over 20 kg): **require two people** or a lifting cart.
- Never lift a printer by its output tray, scanner lid, or paper cassettes — these are not structural.
- Ensure the destination surface is flat, stable, and rated for the weight.

---

# 8.6 Chemical Safety

## 8.6.1 Toner Powder

Toner is a fine plastic powder (polyester or styrene-acrylate) with pigment. While not highly toxic, it requires careful handling.

| Hazard | Precaution |
|---|---|
| Inhalation of fine powder | Work in a well-ventilated area; avoid creating dust clouds |
| Skin irritation | Wear nitrile gloves; wash hands after handling |

| Hazard | Precaution |
|---|---|
| Eye irritation | Wear safety glasses if working with large spills |
| Staining of clothing | Wear a lab coat or apron; toner stains are difficult to remove |
| Fire risk when airborne | Keep toner away from open flames and high heat sources |

### 8.6.2 Toner Spill Response

1. **Do not use a standard vacuum** — it can aerosolize the powder and damage the vacuum motor.
2. Use a toner-specific HEPA vacuum if available.
3. Otherwise, use a dry lint-free cloth to gently lift the powder (do not rub — this grinds it in).
4. For residual dust, use a slightly damp cloth (water only).
5. Allow surfaces to dry completely.
6. Dispose of contaminated materials as electronic waste, not regular trash.

### 8.6.3 Isopropyl Alcohol

- Use in a well-ventilated area — vapors are flammable.
- Keep away from open flames and heat sources.
- Do not spray near the fuser or electrical contacts.
- Store in a sealed container, away from heat and direct sunlight.
- If spilled on skin, wash with soap and water.
- If eye contact occurs, flush with water for 15 minutes and seek medical attention.

### 8.6.4 Cleaning Agents

- Use only cleaners specified in this manual or the manufacturer's guidelines.
- Never mix cleaning products (e.g., bleach + ammonia = toxic chloramine gas).
- Keep all cleaning chemicals in their original, labeled containers.
- Store flammable liquids (isopropyl alcohol) in a fire-resistant cabinet.

---

## 8.7 Personal Protective Equipment (PPE)

### 8.7.1 Required for All Maintenance

- **Nitrile gloves** — for handling toner, dirty components, and chemicals
- **Safety glasses** — when using compressed air or working with toner spills
- **Closed-toe shoes** — in all printer rooms

### 8.7.2 Situation-Specific PPE

| Task | Additional PPE |
|---|---|
| Fuser area maintenance | Heat-resistant gloves (after cool-down), long sleeves |
| Large toner spill | N95 dust mask, safety glasses, gloves |
| Compressed air cleaning | Safety glasses (dust/debris) |
| Moving heavy printer | Steel-toe shoes, back support belt |
| Working above shoulder height | Step ladder (not a chair or box) |

---

## 8.8 Environmental Safety

### 8.8.1 Ventilation

- Printer rooms should have adequate ventilation or HVAC.
- Laser printers emit low levels of ozone and ultrafine particles during operation.
- For high-volume printers (10,000+ pages/month), consider a dedicated ventilation system or air purifier with HEPA and activated carbon.
- Do not place printers in small, unvented closets.

### 8.8.2 Fire Safety

- Keep printers at least 1 meter from flammable materials.
- Ensure the area has a Class C (electrical) fire extinguisher accessible.
- Never use water on an electrical fire — use a $CO_2$ or dry chemical extinguisher.
- If a printer catches fire: **unplug if safe**, evacuate, call emergency services.

### 8.8.3 Waste Disposal

| Waste | Disposal Method |
|---|---|
| Used toner cartridges | Manufacturer recycling program (free mail-back) |
| Waste toner | Seal in plastic bag; e-waste facility |
| Isopropyl alcohol containers | Hazardous waste / chemical disposal |
| Cleaning wipes with toner | E-waste or contaminated waste |
| Old printers / parts | E-waste recycling (do not landfill — contains heavy metals) |
| Paper | Standard paper recycling |

---

# 8.9 Emergency Procedures

## 8.9.1 Electric Shock

1. **Do not touch the victim** if they are still in contact with the power source.
2. Cut the power at the circuit breaker or unplug the device.
3. Call emergency services immediately.
4. If the victim is unresponsive and not breathing, begin CPR if trained.
5. Do not move the victim unless they are in immediate danger.

## 8.9.2 Fire

1. Activate the fire alarm.
2. If safe to do so, unplug the printer.
3. Use a Class C fire extinguisher ($CO_2$ or dry chemical) — **never water**.
4. If the fire is spreading or you cannot control it, evacuate and close doors.
5. Call emergency services.

## 8.9.3 Toner in Eyes

1. Do not rub the eyes.
2. Flush immediately with clean, lukewarm water for at least 15 minutes, holding eyelids open.
3. Seek medical attention.

### 8.9.4 Injury from Moving Parts

1. Power off and unplug the printer immediately.
2. Do not attempt to free a trapped limb by force — disassemble or call for help.
3. Apply first aid for cuts or bruises.
4. Seek medical attention for crush injuries, deep cuts, or suspected fractures.

---

## 8.10 Safety Checklist (Post Every Printer)

Post this checklist near every printer:

- [ ] Power off and unplug before opening covers
- [ ] Wait 30 minutes for fuser to cool
- [ ] Wear gloves and safety glasses when required
- [ ] No water or liquid cleaners near electrical components
- [ ] No standard vacuum for toner spills
- [ ] Do not defeat door interlocks
- [ ] Keep the area clear of clutter and flammables
- [ ] Report all incidents and near-misses

---

*Next: FAQ*

---

# 9 Frequently Asked Questions

## 9.1 General

### 9.1.1 Q: How often should I clean my printer?

**A:** Exterior and scanner glass should be cleaned weekly. Internal components (feed rollers, paper path) should be cleaned monthly. A full deep clean is recommended quarterly. High-volume printers (5,000+ pages/month) may require more frequent cleaning.

### 9.1.2 Q: Can you use third-party (compatible) toner cartridges?

**A:** While third-party toner is less expensive, it carries the following risks: - Print quality might be inconsistent - Toner powder formulation might cause excessive dust and premature wear - Some third-party cartridges might leak and damage the fuser - Using non-OEM supplies might void the printer warranty - Recommendation: use OEM toner for critical printers; test third-party brands on non-critical devices first.

### 9.1.3 Q: How long does a toner cartridge last?

**A:** Toner yield is rated at 5% page coverage (a standard text page). A typical 2,000-page cartridge will print approximately 2,000 text pages, but only 400–800 pages if printing full-page graphics or photos. Actual yield depends heavily on content density.

### 9.1.4 Q: Should I turn the printer off at night?

**A:** It depends on usage: - **Low volume (under 100 pages/day)**: Power off at night to save energy and reduce wear. - **High volume (500+ pages/day)**: Leave on or use sleep mode — frequent power cycling causes more thermal stress on the fuser than continuous low-power operation. - Most modern printers have an auto-sleep / deep-sleep mode that uses less than 5W.

### 9.1.5 Q: What is the ideal room temperature and humidity for a printer?

**A:** 10–32.5 °C (50–90 °F) and 20–80% RH (non-condensing). The sweet spot is 20–25 °C and 40–50% RH. High humidity causes paper to absorb moisture (curl, jams, poor toner fusion); low humidity causes static buildup (multi-feeds, print defects).

---

## 9.2 Print Quality

### 9.2.1 Q: Why are there vertical lines on my printed pages?

**A:** Vertical lines have several causes depending on color and position: - **Black lines**: Damaged drum or toner cartridge. Replace the toner first; if the line persists, replace the drum. - **White lines**: Clogged toner or dirty developer roller. Replace the toner cartridge. - **Lines only on scans/copies**: Dirty scanner glass or ADF glass strip. Clean the glass. - **Repeating marks at regular intervals**: Measure the interval — it matches the circumference of a roller (drum, fuser, or developer). Replace the corresponding component.

### 9.2.2 Q: The toner smudges or rubs off the paper. What's wrong?

**A:** The fuser is not melting the toner properly. Causes include: - Paper is too thick for the current fuser temperature setting (set the correct paper type in the driver) - Fuser unit is worn or failing (replace fuser) - Non-OEM toner with a lower melting point - Print speed is too high for the media weight

### 9.2.3 Q: My color printer has wrong colors. How do I fix it?

**A:** Run the automatic color calibration / registration from the printer's control panel menu. If that does not help: 1. Check that all toner cartridges are genuine and not expired. 2. Replace the cartridge for the missing/incorrect color. 3. Clean the color registration sensors. 4. If banding persists, the transfer belt may need replacement.

### 9.2.4 Q: Why is there a faint duplicate image (ghosting) on my pages?

**A:** Ghosting is caused by a component not fully releasing the toner image. The drum is the most common culprit — replace it first. If ghosting persists, the fuser may be failing. Also check that the paper type setting matches the actual media (thick paper requires higher fuser temperature).

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## 9.3 Paper and Jams

### 9.3.1 Q: Why does my printer keep jamming?

**A:** Recurring jams usually have one of these root causes: 1. **Worn pickup rollers** — the most common cause. Clean with alcohol; if the surface is smooth/glazed, replace them. 2. **Damp paper** — paper absorbs moisture from the air. Store in a sealed package and fan before loading. 3. **Torn paper scrap** — a tiny piece left from a previous jam causes repeated jams. Inspect the entire paper path with a flashlight. 4. **Overloaded cassette** — do not exceed the fill line. 5. **Wrong paper guides** — guides must be snug, not loose or too tight. 6. **Worn separation pad** — causes multi-feeds and jams.

### 9.3.2 Q: Can I use inkjet paper in a laser printer?

**A:** Generally, yes — standard plain paper works in both. However, **inkjet-specific coated paper, photo paper, and transparencies should NOT be used in laser printers**. The laser fuser heat (180–220 °C) can melt inkjet coatings, causing jams and fuser damage. Always check the packaging for "laser compatible" or "laser printer" labeling.

### 9.3.3 Q: Why does the printer pull multiple sheets at once?

**A:** Multi-feeding is caused by: - Static electricity between sheets (fan the paper before loading) - Damp paper (replace with fresh, dry stock) - Worn separation pad (replace it) - Overfilled cassette (reduce paper level) - Paper guides too loose (adjust to touch the stack snugly)

### 9.3.4 Q: Is it okay to print on both sides of previously used paper?

**A:** No. Pre-printed or previously used paper can cause: - Paper jams due to curl or fiber damage - Toner flaking off and contaminating the printer - Ink from the first side melting in the fuser - Misfeeds due to changed paper texture Use only clean, unused paper for laser printing.

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## 9.4 Consumables

### 9.4.1 Q: How do I know when to replace the drum unit?

**A:** Replace the drum when: - The display shows "Drum Life End" or "Replace Drum" - You see repeating spots or smudges at regular intervals (matching the drum circumference, typically 3–10 cm) - Print quality does not improve after replacing the toner - The drum has reached its rated page count (check the configuration page)

**Important:** After replacing the drum, you must reset the drum counter in the printer menu, or it will continue to display the replacement warning.

### 9.4.2 Q: Can I refill toner cartridges yourself?

**A:** Technically yes, but it is not recommended for office environments: - Refilling is messy and toner powder is a respiratory irritant - Refilled cartridges have higher failure rates and leak more often - The drum and wiper blade are not replaced, so print quality degrades with each refill - Most modern cartridges have chips that prevent refilling without chip resetters - For cost savings, consider compatible (new-built) cartridges rather than refills.

### 9.4.3 Q: How should I store spare toner cartridges?

**A:** Store in original packaging, in a cool (10–30 °C), dry (30–70% RH) place, away from direct sunlight and heat sources. Do not remove from packaging until ready to install. Shelf life is typically 2–3 years. Do not store cartridges on their side for long periods — store upright as indicated on the packaging.

### 9.4.4 Q: What's the difference between a maintenance kit and a fuser?

**A:** A maintenance kit is a bundle of wear items that includes the fuser, transfer roller, pickup rollers, and separation pads. It is replaced at a set page interval (typically 100,000–200,000 pages). The fuser alone is replaced only if it fails before the maintenance interval. Always replace the full kit when due — replacing only the fuser leaves worn rollers that will cause jams.

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## 9.5 Network and Software

### 9.5.1 Q: The printer shows "Offline" but it is powered on. How do I fix it?

**A:** Try these steps in order: 1. Print a configuration page from the printer and note its IP address. 2. Ping the IP address from a computer. If it fails, the printer has a network issue (check cable, switch port, DHCP). 3. If ping succeeds, the issue is on the computer side. Restart the print spooler (`net stop spooler` then `net start spooler`). 4. Remove and re-add the printer using the correct driver. 5. Ensure the printer has a **static IP address** — DHCP addresses change and break the connection.

### 9.5.2 Q: Can I print from my phone or tablet?

**A:** Yes, if the printer supports it: - **Apple AirPrint**: Most modern printers support this natively — no app needed. - **Google Cloud Print**: Deprecated as of 2020; use manufacturer apps instead. - **Manufacturer apps**: HP Smart, Canon PRINT, Brother iPrint&Scan, etc. - **Mopria**: Android standard for printing to Mopria-certified printers. Ensure the mobile device and printer are on the same Wi-Fi network.

### 9.5.3 Q: How do I clear a stuck print job?

**A:** On Windows: 1. Open **Settings > Devices > Printers & scanners** 2. Select the printer > **Open queue** 3. Click **Printer > Cancel All Documents** 4. If jobs remain stuck, restart the print spooler service: - Press `Win + R`, type `services.msc` - Find **Print Spooler**, right-click > **Restart**

On macOS: System Settings > Printers & Scanners > select printer > Open Print Queue > delete jobs.

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## 9.6 Safety

### 9.6.1 Q: Is toner toxic?

**A:** Toner powder is generally low in toxicity, but it is a respiratory irritant due to its fine particle size. It can cause eye, skin, and respiratory irritation with heavy exposure. Always: - Wear gloves when handling cartridges - Avoid creating toner dust clouds - Use a toner-specific HEPA vacuum for spills (never a standard vacuum) - Wash hands thoroughly after handling - If toner gets in eyes, flush with water for 15 minutes and seek medical attention

### 9.6.2 Q: The printer smells like ozone. Is that dangerous?

**A:** A faint ozone smell during long print runs is normal for laser printers. However: - Strong or persistent ozone odor may indicate a failing corona wire or ozone filter - Ensure the room is well-ventilated - Replace the ozone filter per the maintenance schedule

**Smoke or burning odor**

If the smell is accompanied by smoke or burning odor, power off immediately and service the printer.

### 9.6.3 Q: Can I open the fuser to clean it?

**A:** No. The fuser is a sealed, high-voltage, high-temperature assembly. Opening it: - Risks severe burns (fuser runs at 180–220 °C) - Risks electric shock (high-voltage contacts) - Can damage the delicate fuser roller coating - May void the warranty Fuser cleaning should be limited to wiping the entrance/exit guides with a dry cloth after full cool-down. Internal fuser service requires a qualified technician.

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## 9.7 Cost and Efficiency

### 9.7.1 Q: How can I reduce printing costs?

**A:** Effective strategies include: 1. **Duplex printing** — set double-sided as the default (cuts paper cost by ~50%) 2. **Draft/Toner Save mode** — for internal documents, use lower resolution and toner save 3. **Black and white default** — set monochrome as default; color only when needed 4. **Print preview** — always preview before printing to avoid wasted pages 5. **Digital workflows** — use scan-to-email/PDF instead of printing and distributing 6. **Managed print services** — for large fleets, an MPS contract can reduce costs by 20–30% 7. **Right-size the fleet** — eliminate underutilized printers; consolidate to shared multifunction devices

### 9.7.2 Q: What is the typical lifespan of an office printer?

**A:** - **Personal / small office printers** (under 30 pages/min): 3–5 years or 100,000–200,000 pages - **Workgroup printers** (30–50 ppm): 5–7 years or 500,000–1,000,000 pages - **Enterprise multifunction printers** (50+ ppm): 7–10 years or 1–5 million pages Lifespan depends heavily on usage volume, maintenance quality, and environmental conditions. A well-maintained printer can exceed these estimates; a neglected one may fail early.

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*Return to Home*

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# 10 Glossary

## 10.1 Overview

This glossary defines the terms used throughout the Printer Maintenance Manual. It is written for designated printer operators, IT support staff, and anyone who completes a service log entry.

Definitions describe general office printing equipment. Model-specific part names, replacement intervals, and error codes are documented in the manufacturer's service manual, which always takes precedence over this guide. Terms are grouped by topic so that related concepts appear together. Shortened forms are listed in the Abbreviations section.

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## 10.2 Hardware Components

### 10.2.1 Imaging and Fusing

| Term | Definition |
|---|---|
| **Drum unit** | Photosensitive cylinder that receives the latent image and transfers toner to paper. On many laser printers it is a separate, replaceable consumable. |
| **Fuser unit** | Heated roller assembly that melts toner into the paper fibers under pressure. It reaches high operating temperatures and must cool fully before any maintenance. |
| **Transfer belt** | Belt that carries the toner image from the imaging drums to the paper on color laser printers. |
| **Transfer roller** | Roller that applies the charge that pulls toner from the drum onto the paper. |
| **Corona wire / charge roller** | Component that applies a uniform electrostatic charge to the drum before exposure. A dirty or failing one causes streaks and uneven density. |
| **Laser scanner unit** | Optical assembly that writes the latent image onto the drum with a laser beam. Dust on its window produces faded or repeating defects. |
| **Developer roller** | Roller that delivers toner to the drum during development. |
| **Ozone filter** | Filter that captures ozone produced by high-voltage charging components. A strong, persistent odor can indicate a filter or corona wire problem. |

| Term | Definition |
|---|---|
| **Waste toner container** | Reservoir that collects surplus toner removed from the drum during cleaning. Replace it before it fills. |

### 10.2.2 Paper Path and Feeding

| Term | Definition |
|---|---|
| **Pickup roller** | Roller that lifts the top sheet from the cassette and feeds it into the paper path. Worn or glazed rollers are the most common cause of misfeeds. |
| **Separation pad** | Friction pad that holds back all but the top sheet, preventing multiple sheets from feeding at once. |
| **Feed roller** | General term for the rollers that move paper along the path between the cassette and the output tray. |
| **Registration** | The alignment of the paper and of each color layer. Poor registration produces skewed sheets or color fringing. |
| **Duplex unit** | Mechanism that turns the sheet over for two-sided printing. |
| **Automatic document feeder (ADF)** | Tray that feeds originals across the scanner glass for copying or scanning, so that pages do not need to be placed one at a time. |
| **Output tray** | Bin that receives finished pages. An overfull tray can cause jams at the exit. |
| **Bypass tray / multi-purpose tray** | Manual feed tray for single sheets, envelopes, and non-standard media. Keep it free of dust and close it fully to avoid false jam reports. |
| **Finisher / staple unit** | Optional output accessory that sorts, staples, or hole-punches completed sets. Refill staples and empty the trim tray before they fill, or jobs will stop partway through. |

## 10.3 Consumables and Supplies

| Term | Definition |
|---|---|
| **Consumable** | Any item that is used up during normal operation and replaced on a schedule, such as toner, ink, drums, and waste toner containers. |
| **OEM supplies** | Cartridges and parts made by the printer manufacturer. Compatible (third-party) supplies are new-built alternatives; remanufactured or refilled cartridges are neither. |
| **Toner yield** | Estimated page count for a cartridge, normally rated at 5% coverage of a standard page. Dense graphics reduce the actual yield. |
| **Maintenance kit** | Bundled set of wear items, typically a fuser, transfer roller, pickup rollers, and separation pads, replaced together at a set page interval. |
| **Meter reading** | Total page count reported by the printer. Record it daily to track usage and plan preventive maintenance. |
| **Shelf life** | Period an unopened consumable remains within specification when stored as directed. |

**Yield is an estimate**

A cartridge rated for 2,000 pages prints far fewer pages when jobs contain full-page graphics or photos. Use the meter reading and observed print quality, not the rated yield alone, to plan replacements.

## 10.4 Print Quality Terms

| Term | Definition |
|---|---|
| **Ghosting** | Faint duplicate of an image appearing further down the page, usually caused by a drum or fuser that does not fully release toner. |
| **Banding** | Repeating light or dark bands across the page, often linked to the transfer belt or color calibration. |
| **Streak** | Continuous line running in the feed direction. Black streaks commonly point to the drum or cartridge; white streaks to a toner clog. |
| **Repeating defect** | Mark that recurs at a fixed interval. The interval matches the circumference of the faulty roller, which helps identify the component. |

| Term | Definition |
|---|---|
| **Fading** | Overall light or uneven print density, commonly caused by low toner, damp paper, or toner save mode. |
| **Smudging** | Toner that rubs off after printing, indicating that the fuser did not bond the toner to the paper. |
| **Color calibration** | Control panel procedure that aligns color layers and corrects density. Run it before replacing hardware for color complaints. |

## 10.5 Paper and Media Terms

| Term | Definition |
|---|---|
| **Media** | Any material that passes through the printer, including plain paper, envelopes, labels, and transparencies. |
| **Basis weight** | Paper weight expressed in g/m² (gsm). Media outside the printer's supported range causes jams and poor fusing. |
| **Grain direction** | Direction in which most paper fibers align. It affects curl and how smoothly a sheet travels through the path. |
| **Double feed** | Two or more sheets entering the paper path together, caused by static, damp stock, a worn separation pad, or an overfilled cassette. |
| **Misfeed** | Failure to pick up a sheet, or a sheet that enters at an angle. |
| **Jam** | Sheet stopped inside the paper path. Always check for torn scraps after clearing one. |
| **Laser-safe media** | Media rated to withstand fuser temperatures. Inkjet-coated and photo media can melt inside a laser printer. |

## 10.6 Network and Software Terms

| Term | Definition |
|---|---|
| **Print driver** | Software that converts a document into the printer's page description language and sends it to the device. |
| **Print spooler** | Operating system service that queues print jobs. Restarting it clears jobs that remain stuck after cancellation. |
| **Print queue** | Ordered list of jobs waiting for a printer, visible from the operating system. |
| **Static IP address** | Fixed network address assigned to the printer. A changing DHCP address breaks driver connections and makes the printer appear offline. |
| **Configuration page** | Report printed from the control panel that lists the IP address, firmware version, installed options, and supply status. |
| **Firmware** | Software embedded in the printer. Update it on a planned schedule, not during business hours. |
| **Scan-to-email / scan-to-folder** | Functions that deliver scanned documents to a mailbox or network share without a connected computer. |
| **Page description language** | Format used to describe a page to the printer, such as PCL or PostScript. A mismatched driver can produce garbled output. |

## 10.7 Maintenance and Service Terms

| Term | Definition |
|---|---|
| **Preventive maintenance** | Scheduled work performed before a failure occurs, such as deep cleaning and maintenance kit replacement. |
| **Service log** | Record of checks, meter readings, replaced parts, and incidents for a single printer. See the template in Home. |
| **Asset record** | Inventory entry that identifies a printer by model, serial number, location, and warranty or contract status. |

| Term | Definition |
|---|---|
| **Escalation** | Handover from routine operator maintenance to IT support or a qualified technician. See Troubleshooting Guide. |
| **Service contract** | Agreement that defines response times, covered parts, and on-site support. Verify coverage before ordering parts. |
| **Qualified technician** | Person trained to service high-voltage or high-temperature assemblies. Fuser and internal lubrication work is restricted to this role. |

**Terminology does not replace safety rules**

Several entries above refer to high-voltage or high-temperature assemblies. Read Safety Guidelines before opening any cover, and never service a printer that is powered on.

## 10.8 Abbreviations

| Abbreviation | Full form |
|---|---|
| **ADF** | Automatic document feeder |
| **DHCP** | Dynamic Host Configuration Protocol |
| **DPI** | Dots per inch |
| **GSM** | Grams per square meter |
| **IP** | Internet Protocol |
| **LSU** | Laser scanner unit |
| **MFP** | Multifunction printer |
| **OEM** | Original equipment manufacturer |
| **PCL** | Printer Command Language |

| Abbreviation | Full form |
|---|---|
| **PPM** | Pages per minute |
| **PPE** | Personal protective equipment |
| **RH** | Relative humidity |
| **SNMP** | Simple Network Management Protocol |
| **USB** | Universal Serial Bus |

## 10.9 Related Pages

- Getting Started — printer components and tools
- Daily Checklist — scheduled checks and meter readings
- Consumables Management — replacement and inventory procedures
- Troubleshooting Guide — symptom-to-solution reference
- Safety Guidelines — hazards and protective equipment

*Return to Home*